PPPE [redacted]

Group C Final Paper - Purview of A Grid Integration Scheme

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The University of Texas at Dallas

Background and Literature Overview

In developed political entities, it is hard to underestimate the importance of electricity. The provision of this essential utility ranks in necessity as one scarcely appearing less than that of water. Yet, ideas on how best to supply this resource have not been such that the word 'consensus' can yet lay a claim to describe them. Specifically in the populous state of Texas, the intersection of the unique attributes of electric power distribution (Cohn 2022) and the changing demographic realities of the people that compose it (Melhado 2023, Wilder 2023) have combined to create some friction in regard to the best means of literally powering the Lone Star state in the future. Especially have opinions centered upon whether Texas should maintain its current independent means of operating its electricity grid separately from all other power systems in continental North America or continue on its path of relative 'independence' in this regard (Oxner 2022, Wagner 2024).

We will propose in this paper a novel admixture of analysis and conceptual modeling that we hope to demonstrate can provide policymakers, citizens, and other stakeholders with another avenue with which to consider the possible trajectories of power transmission in Texas. More concretely, we will attempt to provide an assessment of the feasibility and likely impacts of the scenario wherein the sovereign nation of Mexico and the state of Texas join their respective independent power grids into a single, synchronized power distribution network and manage it under the Texas 'ERCOT style' of electricity provision. We will analyze this merging along the reliability dimension of electricity provision. We chose to focus on the ERCOT system especially because it exhibits certain economic characteristics (Baldick et al.) that we will argue make it amenable to economic systems assessment of the type we will undertake in this paper.

Further, we will contextualize this assessment with regard to the possibilities inherent in the state of Texas joining with other major power grids in operation across the U.S. (a move that has the support of some lawmakers (Wagner 2024)). We will also draw parallels to and insights from the synchronous electric grid currently operating across most of Western Europe and Turkey.

While certain asynchronous 'tie-ins' currently connect the ERCOT grid to the Mexican grid (Proffer 2024), we are proposing a wholly novel combination that could better be thought of as a 'tie-in' at the system level. This scenario, of course, is perhaps best conceptualized as a sort of thought experiment. The conditions necessary to bring about such a monumentous restructuring across national lines could arguably come about only through fairly unprecedented evolutionary trajectories and/or socio-political change. Yet, we will argue that the European example may provide some grounding as to how such a scale of integration may come about (Roques 2021).

To reiterate, we seek to discuss and ascertain whether a full system integration between the electric grids of both the state of Texas and the nation of México can more reliably provide real-time electricity coverage to constituents of Texas and provide a model of the true (economic) trade-offs of such an integration. To our knowledge, this proposition has never been assessed before in the literature. To be conducted most accurately such an assessment would require that the dynamic impacts that various physical, economic, and political constraints that would simultaneously operate under such a system be incorporated within it. However, the specificity of such incorporation no doubt will hinge upon any underlying assumptions taken and the level of interactivity that is of interest to the analysis. While a realistic attempt to wholistically model such a scenario at nearly any level of granularity would no doubt require

some of the most sophisticated tools currently in use in either energy or systems research (such as those discussed in Birchfield et al., 2023), we hope to be able to provide a relatively static but no less grounded theoretical basis from which to ascertain the broad outlines of such a hypothetical system merger. Our overarching research design will analyze the impact of a complete merger mostly under the purview of standard welfare utility theory. It will also hopefully provide a starting point for further, more quantitative, research in the future.

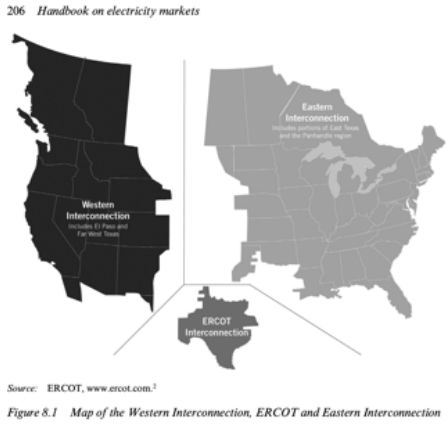
Methodology

"Join the Union?"

As a first point of contact, it might seem natural to ask whether a full-scale integration could not be achieved between Texas's grid and that of the other two main grids servicing the rest of the United States, namely the Western Connection and the Eastern Connection (Baldick et al. 2021, 206). Indeed, this proposal has been advanced before in spheres other than the purely academic, most recently by lawmakers such as Congressional Representatives Greg Casar and Alexandria Ocasio-Cortez when they pushed their "Connect the Grid Act", which would force the ERCOT system to integrate with these other systems (Wagner 2024).

Such an integration would 'unsituate' the unique borders of the ERCOT Independent System Operator (ISO) that currently is responsible for providing electricity to approximately 90% of all Texans (Bladick et al. 2021, 205). This hypothetical coupling would also directly connect it with the grids currently maintained by the nine other ISOs/ Regional Transmission Organizations (RTOs) that provide electricity in the U.S. and that operate within either the Western or Eastern Interconnection (IRC 2024). For purposes of simplicity, we can assume that electricity in the system is transmitted under the aegis PJM-style market design, which broadly forms the basis of the market operation of all nine ISOs/RTOs (Hogan 2021, 182-183). For the

purposes of this section, that only amounts to recognizing that wholesale electricity costs are largely determined as they are *on average* in the U.S. – i.e. by market forces (2021, 182).

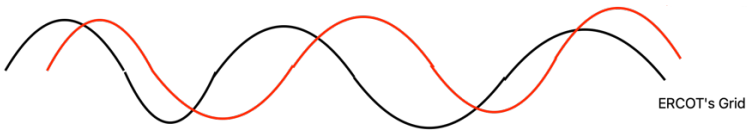


Map of U.S. grids, from Baldick et al. 2021, 206

From a strictly engineering point of view, there are two properties physical law demands an electrical grid must possess in order to run *synchronously* (or all together at once): at high voltage levels (that which transpires from generation in high-rise transmission wires) all directly connected units must run at a single frequency and in sinusoidal phase, also known as system stability (Blume 2017, 165-177). In most complete grids, such as the Western Connection or ERCOT, the high-wire transmission lines meet this standard in a single grid, though step-down transformers allow changes in voltage to be 'wound down' before power is delivered to most residential and commercial end users (2017, 45).

Indeed, the way most generation units are brought 'in sync' with electric grids *now* is basically via a process of 'tuning' them until their frequency and phase match that of the wider grid they are being 'paired' to (Csanyi 2013, 2015 and 2023). Feasibly, then, syncing high power lines across currently existing physical transmission lines in all three girds is possible, provided transmission lines directly connect them together. It just must be kept in mind this requires the

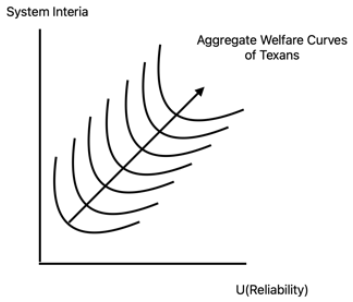
entire combined system (and all generation units within it) to be maintained at the same phase and frequency continuously.



Hypothetical sine waves of ERCOT and other grid(s)

Of what direct benefit would Texan residents receive by such a join? Adding interconnections to a utility grid typically increases system reliability – this property is an attribute of inertia, which essentially means the ability of a grid to get back in sync when any part of a grid becomes 'unsynced' with the others (Blume 2017, 172). The more rotating machine generators (2017, 172) that are connected in such a system, the more inertia it *should* have. A synced integration would theoretically increase the inertia of a combined power system by as much as the amount of rotating-machine-based generation capacity each grid could bring to the system.

Essentially, the benefits will boil down to making electricity provision more reliable for Texans. Most saliently, this increase in system inertia *should* result in greater productivity. Leading to greater aggregate utility derived per dollar of energy input across all industries, *all else held equal* due to more consumable output per given unit of input – this can be considered a parallel case to a positive income effect which should increase utility because all goods become relatively cheaper to ultimately consume (Autor 2004, 1).



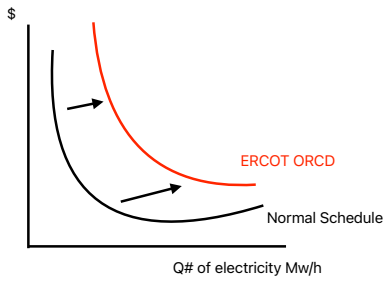
Simplified diagram of utility increase due to merger. More output is produced with a given dollar of input as inertia increases.

Putting a number value on the value of any increase in reliably is difficult. Namely, because such an integration scenario would be a dynamic one that might be highly sensitive to initial state 'tuning parameters' (Ormand 2024, Pagani and Aiello 2012) of the current generation mix, infrastructure, and the unforeseeable changes in short-term demand that such a transformation may entail. While a precise quantitative assessment of such savings would no doubt be desirable, if it can be presumed that such stability could at least afterward prevent another statewide catastrophe such as that seen in the aftermath of winter storm Uri, a possible lower boundary of $56.8 billion dollars (via Perryman 2021) in benefit seems acceptable.

Texas and Mexico in Sync

The incorporation of the grids of Texas and Mexico into a single, synchronous grid is also physically possible in the manner that such a union is possible between Texas and the other grids in operation in the rest of the United States. This is because under the auspices of the *National Power System* (*SEN*), the nation of Mexico possesses modern electricity generation, transmission, and distribution infrastructure (ITA, EIA) that could theoretically be 'synced' with the ERCOT system via the two physical requirements of matching system-wide frequency and phase as discussed above. It would simply necessitate that Texas 'switches' to Mexico's numbers in this regard, or vice versa. On the economic side, we will presume this proposed merged system operates under the market provision structure that currently dominates within the ERCOT ISO. We do this primarily because it appears most likely that such a system would be economically efficient, for reasons to be discussed shortly, and therefore represents a sort of 'best case' model of electricity provision.

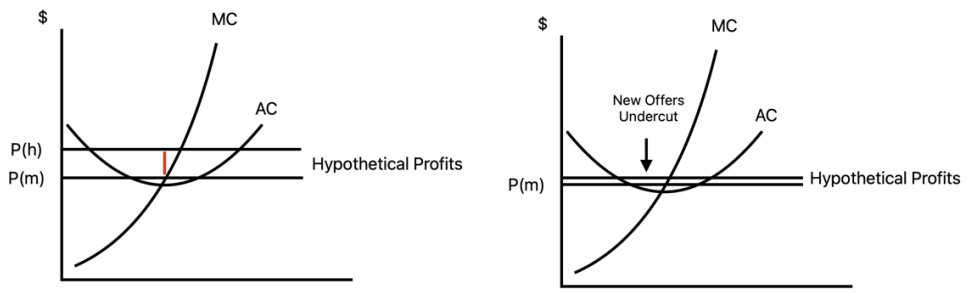
Perhaps the most salient feature of the ERCOT energy market is the 'energy-only' bias upon which the ERCOT wholesale market operates (Baldick et al. 2021, 221-223). This feature necessitates that there are no official reserve requirements in place at the state level and that when overall system capacity drops below a certain level, the market begins to function off of an operating reserve-demand curve or ORCD - this is the crucial rationing mechanism that implements the market-oriented vision of the system (2021, 223-224, 228-229) beyond more normal operating parameters, which can allow (wholesale) electricity prices to reach up to $9,000 per hour (2021, 231-232). As prices infamously did in the state during winter storm Uri in 2021 (Busby et al. 2021, 4).



ORCD Demand Curve

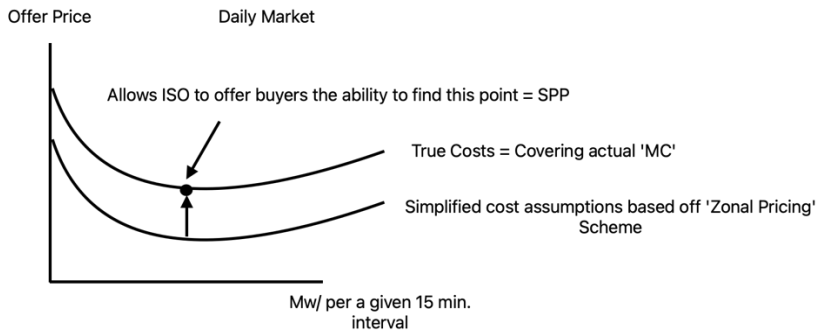
Although fallbacks to ISO-induced black-out are still resorted to in emergencies (Blume 2017, 169, 181-182), it is a purposeful design of the Texas system to take market mechanisms seriously as the best arbiter of investment and consumption decisions by energy providers and consumers (Baldick et al. 2021, 213, 237). Indeed, at nearly any scale except the seconds-scale range, at which direct ISO supervision is required due to the simultaneous balance of supply and demand that must be kept in order to keep electric grids running (2021, 222), some economists have argued that a market-centric approach such as ERCOT's should lead to an efficient provision of electricity (Wolak 2021). Mainly because a market system, at least concurrent with

certain conditions such as the existence of fixed-price forward contracts (Wolak 2021, 88-89, 107), is able to steer suppliers to offer provision at close to the marginal cost of supplying their energy because attempting to over-charge would mean other firms could undercut them and 'capture' their profits (2021, 89, Baldick et al. 2021, 219, 225). Not all economists would agree with the virtues of market efficiency in regard to electricity provision, however (Borenstein et al. 2023, 182).



Simplified diagram of competitive effects

Another key mechanism of the ERCOT setup is the utilization of nodal pricing or LMP (locational marginal pricing). This pricing scheme was implemented in Texas after numerous delays in 2010 (Baldick et al 2021, 208), following other wholesale privatization efforts that began in the 1990s (Cohn 2022). It allows prices to implicitly incorporate information about infrastructural limitations and real generation and capacity constraints, and therefore, ISOs can factor them into a workable lowest-cost 'mix matrix' given offers by suppliers to meet overall system demand (Wolak 2021, 79-80, Baldick et al. 2021, 208-209, 213, 218).



Simplified diagram of nodal-lead pricing efficiency

It is such a system that situates and provides the substrate for the day-ahead bidding structure of the ERCOT market (Baldick et al. 2021, 215-216). With such a setup, a settlement point price (SSP) is determined that then informs pricing plans for consumers (Texas Grid Analytics). Ideally, then, a merged grid system operating under such a regime should be able to provide (wholesale) electricity prices at economically efficient levels.

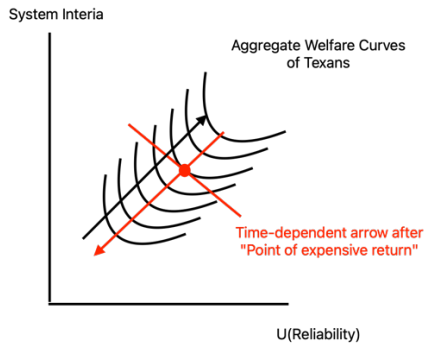
It is worth pointing out that not all researchers agree on the efficacy of the ERCOT system in providing the best possible set of trade-offs when it comes to meeting desiderium such as infrastructural investment or equitable provision. On the latter front, Tiwari (2022, 6-7) has argued that the ERCOT system fails to adequately consider the impact the limitations of the pricing system have on the most vulnerable Texans.

In regard to infrastructure, Busby et al. (2022, 7) argue that the current Texas grid needs further weatherproofing installed in order to buttress against severe weather events. A point echoed by Liu et al. (2024, 3-4) when they concluded that Texans should increase the resiliency of their grid after they concluded outages associated with winter storm Uri cost Texans a 1.44% loss in GDP. Levin et al. (2022, 2-3) note that under-investment in longer-term grid performance measures may be attributable to their relative unattractiveness to suppliers in a market-based system such as ERCOT's because such investments may be perceived as too high risk. The risks

of underinvestment are further echoed by several economists at the Dallas Federal Reserve, who conclude that the total cost of winter storm Uri might have exceeded $100 billion and that weather-proofing measures should be implemented (Golding et al. 2021). Although some commentators think market-based solutions could still remedy some of these deficiencies (Clemenhagen 2022).

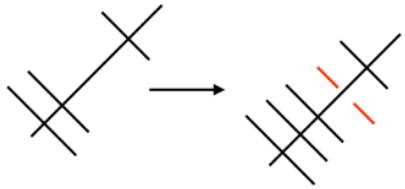
Finally, there are some potential drawbacks at the system level of such a merger. The increase in system inertia that is supposed to 'fuel' increased reliability can only be increased via traditional rotating thermo-machines such as steam turbines and not via renewable energy sources such as unmechanical solar panels (Ayamolowo et al. 2022, 4). Wind turbines also might not be able to supply as much inertia to a grid they are connected to as more traditional energy generation units (Abreu and Shahidehpour 2006, Ayamolowo et al. 2022, 2). However, it should be pointed out that some researchers believe this limitation can be overcome given certain configurations of specific turbines to specific grid settings (Muljadi et al. 2012).

This is pertinent, as both Texas and Mexico wish to increase the proportion of their energy input mixes that are supplied by renewable sources in the future (Wamsted 2024, ITA). This means that if the renewable energy source mix is to increase, the benefit that this will bring will have to be traded off against the stabilizing benefits of a hypothetical merged 'super grid' that would be lost with traditional thermo-machines when they are eventually replaced. This may even result in lower overall welfare (utility) if the tradeoff for 'renewables later' costs more than any gain in 'inertia' that was ever received. This is especially pertinent in a world that remains skeptical of embracing one feasible renewable source of energy generation that could theoretically still provide the system with a decent inertia boost due to their rotating steam turbines: fission-based nuclear plants (Saad 2022, Ayamolowo et al. 2022, 11).



Graph of utility going down after renewables must abrogate inertia gains

In addition, electric grids can also suffer from a phenomenon known as Braess's Paradox. Essentially, this is a property of complex systems such that after a certain point, the more connections that there are in that complex network, the more 'unstable' that network can become (Schäfer et al. 2022, 1-2). This has been shown to be a potential source of problems for the operation of electric grids in particular (Schäfer et al. 2022). These system-level effects could, of course, in principle affect either the Texas-Mexico merger or the Texas-U.S. merger.

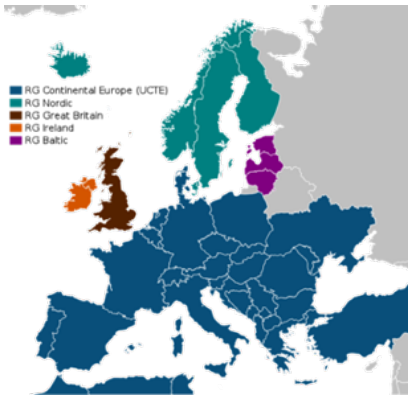


Hypothetical system breaking via Braess's Paradox

A European Style Union?

How might the structure of such an integrated Tex-Mex system compare to that currently prevalent in Europe? Can it address some of the system-level concerns that the purely theoretical analysis considered above? The current architecture of the European grid system is one that, on a *prima facie* basis, closely resembles an actualized version of the hypothetical super-merger between Texas and Mexico under consideration in this paper. Currently, the grid connecting continental Europe is the largest *synchronous* grid in operation in the world (Liu et al. 2020),

crossing political boundaries from Spain, France, Germany, Italy, to parts of Ukraine and Turkey (below). The current market structure of this grid, like the Texas case, also resulted from liberalization efforts to implement free-trade practices in the 1990s (Roques 2021, 308, 310-311).



Map of European continental grid (blue), copyright Wikimedia Commons 2022

However, while Europe seems to have successfully resolved grid integration issues on a physical system level, on a purely provisional basis, there is much less harmony apparent. Fabien Roques has argued that, since the turn of the century at least, the market for electricity generation and transmission has become fragmented due to technological changes and various differences in governance priorities (2021, 320-321, 327). He primarily blames this on the lack of adoption of an integrating market design by the EU at the outset of market-based reforms (2021, 327).

Potentially, in our hypothetical Tex-Mex system, the systemwide adoption of the ERCOT scheme would mitigate this issue. Although it may remain a pertinent one depending on the terms under which Texas was to join a strictly U.S.-based grid. Even in the Tex-Mex example, however, it seems reasonable to suppose that it is likely that frictions in cultural, business, or bureaucratic practices across different nations not used to such cooperation may become an obstacle to the smooth operation of a single-system electric grid.

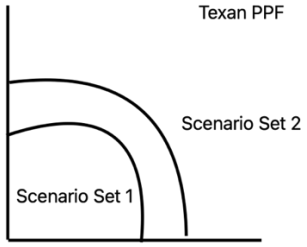
Yet, while many other benefits may have accrued to Europe as a result of integration so far, namely the increase in cross-border trade to electricity (Roques 2021, 314-316), it should be pointed out that some researchers have found that such greater cross-border flow has not resulted in lower prices for consumers (Klopčič et al. 2022, 7-8). In addition, some have noted that the commitment structure imposed upon electricity suppliers in Europe may inadvertently allow the exercise of market power by owners of generation units in at least certain circumstances (Wolak 2021, 80). Although, even the ERCOT system is not always immune from this potentiality (Baldcik et al. 2021, 217). Yet it may seem that for the European case, an argument can be sustained that it does not represent the best implementation of either the PJM or ERCOT-based market visions of the two hypothetical merger scenarios considered above.

All these factors considered, then, may make the European example less of an ideal model for refracting hypothetical 'merger constructs' off of than may at first be supposed. What may well be concluded, however, is that there is a need for caution. If two polities (of whatever strata, be it state, region, nation, etc.) wish to integrate something as essential as their electric grids, it seems advisable that this should be done in a manner that cogently takes account of the various socio-economic goals of each society. This includes environmental and wider policy concerns in addition to purely (price-centric) economic ones.

Results and Concluding Remarks

On this understanding, it may be advised for a generally risk-averse society to prefer an electricity provision enhancing option other than the potentially uncertain, multi-dimensional, massive integrations of the scale considered here. Notably, this may be most easily achieved by expanding the number of indirect, DC, tie-ins that connect two or more electric grids. Indeed, by one estimate the inclusion of a single tie-in may spare Texans from a future repeat of failure on

the scale of winter storm Uri in the future at a cost of only 2.6 billion (Proffer 2024). This expansion might even be 'paid for' by the 'savings' in transmission costs that could eventually transpire from incentivizing greater renewable generation capacity in the state (Woo et al. 2020). Considering the estimated scale of damages that could, therefore, be prevented (see above), that might be a relative bargain for productivity increases.



Hypothetical Texan Production Possibilities Frontier (PPF) rising with new DC tie-in(s)

While we do not conclude with an unqualified recommendation for a Texas-Mexico 'super grid,' we believe that the ultimate decision must be grounded in the risk profiles and socio-political context of the political entities under consideration. We hope our conceptual analysis can provide some grounding for a more extensive understanding in that regard.

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